

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I __G.ALTHAF(192372004)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __G.ALTHAF_____

Criteria	Relevance to Application (1.25)	Use of Cloud Concepts (1.25)	Practical Insight (1)	Completeness (0.75)	Clarity (0.75)	TOTAL (5)
Weightage	25%	25%	20%	15%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Application Based Questions

1. How does cloud federation improve resource sharing and availability across datacenters?
2. What are the key security issues in virtualization, and how can they be mitigated?
3. How does a software-defined datacenter enhance scalability and management?
4. How can identity and access control ensure data privacy in cloud-based healthcare systems?
5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Solutions:-

Question 1: Cloud Federation Resource Sharing & Availability

1. Relevance to Application

- **Application Scenario:** Multi-national enterprises or global web applications operating across distinct geographical regions that experience localized traffic spikes or strict local compliance laws.
- **Relevance:** Cloud federation allows distinct cloud providers or isolated datacenters to interconnect their infrastructures, forming an aggregate, collaborative computing environment.

2. Use of Cloud Concepts

- **Interoperability and Standard APIs:** Utilizing uniform communication protocols to bind disparate cloud providers together transparently.
- **Dynamic Resource Orchestration:** Managing structural scale across heterogeneous environments via central cloud brokers.
- **Geographical Redundancy:** Distributing virtualized application deployments globally to drastically reduce network latency and optimize regional content delivery.

3. Practical Insight

- **Handling Surge Capacity (Cloud Bursting):** When a local datacenter experiences a massive spike in user traffic that exhausts local compute limits, the federation mechanism automatically pushes overflow workloads to an interconnected secondary datacenter, preventing service degradation.
- **High Availability & Disaster Recovery:** If a catastrophic infrastructure failure occurs at one datacenter, traffic is automatically re-routed to a live, mirrored partner datacenter within the federation, maintaining near-zero downtime.

4. Completeness & Clarity

- **Conclusion:** Cloud federation effectively prevents vendor lock-in and resource stagnation by turning siloed datacenters into a fluid, highly available global resource pool.

Question 2: Virtualization Security Issues & Mitigations

1. Relevance to Application

- **Application Scenario:** Multi-tenant public or private cloud environments where workloads belonging to completely separate organizations share the same underlying hardware platform.
- **Relevance:** Virtualization fundamentally alters the security boundary from physical components to software abstractions, creating unique vectors for target exploits.

2. Use of Cloud Concepts

- **Hypervisor Vulnerabilities:** The hypervisor represents a single point of failure; a compromise here compromises all hosted guest VMs.
- **Side-Channel Exploits:** Attacks targeting CPU architectures (e.g., Specter or Meltdown) where malicious VMs deduce sensitive cryptographic data from neighboring virtual spaces.
- **VM Escape:** An attack vector where code executing inside a guest VM bypasses the virtual machine monitor layer to run unauthorized commands directly on the host OS.

3. Practical Insight

- **Mitigating VM Escape & Hypervisor Vulnerabilities:** Cloud administrators must enforce rigorous, routine hypervisor patching schedules alongside deploying specialized bare-metal architectures that separate management networks from tenant traffic.
- **Mitigating Multi-Tenancy Threats:** Enforcing strict CPU pinning, disabling simultaneous multithreading (SMT) where high-security workloads dictate, and implementing virtualized micro-segmentation at the network layer to block unauthorized lateral traffic between adjacent VMs.

4. Completeness & Clarity

- **Conclusion:** Securing virtualized environments requires an ongoing strategy of defense-in-depth, treating the hypervisor as the most critical infrastructure asset.

Question 3: Software-Defined Datacenter (SDDC) Scalability & Management

1. Relevance to Application

- **Application Scenario:** Enterprise infrastructure scaling rapidly to meet shifting consumer market demands without requiring manual physical hardware modifications every time a new resource is deployed.
- **Relevance:** An SDDC abstracts, pools, and automates all foundational datacenter elements—compute, storage, and networking—into comprehensive software logic.

2. Use of Cloud Concepts

- **Infrastructure as Code (IaC):** Defining, provisioning, and maintaining complex architectures using declarative configuration scripts (e.g., Terraform).
- **Abstracted Resource Pooling:** Aggregating physical server arrays into singular, elastic virtual clusters.
- **Centralized Control Plane:** Managing the entirety of the infrastructure lifecycle through unified software dashboards or API integration points.

3. Practical Insight

- **Automated Scalability:** When an application requires additional resources, the SDDC software automatically provisions the necessary virtual machines, allocates virtualized storage, and updates network switches instantly without human administrative intervention.
- **Simplified Lifecycle Management:** Hardware maintenance becomes trivial; administrators can seamlessly put a physical server into "maintenance mode," prompting the SDDC to automatically live-migrate all operating workloads onto alternative hardware with zero application interruption.

4. Completeness & Clarity

- **Conclusion:** SDDCs shift the operational focus away from error-prone manual hardware management into predictable, high-speed software automation.

Question 4: Cloud-Based Healthcare Identity & Data Privacy

1. Relevance to Application

- **Application Scenario:** Hospital networks storing highly protected electronic health records (EHRs) in a cloud infrastructure while requiring instant data access for doctors, nurses, and external laboratories.
- **Relevance:** Healthcare architectures must strictly satisfy compliance regulations (like HIPAA or GDPR) to shield sensitive clinical data from unauthorized viewing while remaining highly accessible.

2. Use of Cloud Concepts

- **Role-Based & Attribute-Based Access Control (RBAC/ABAC):** Defining context-driven rules that calculate dynamic data access configurations.
- **Data-at-Rest and Data-in-Transit Encryption:** Safeguarding medical records using cryptographic standards managed by Cloud Key Management Services (KMS).
- **Comprehensive Audit Trails:** Maintaining unalterable, automated logs detailing precisely who accessed or modified patient data.

3. Practical Insight

- **Implementing Zero-Trust & Privacy Policies:** Utilizing ABAC rules ensures that a user can only view a specific patient record if they are the designated attending physician *and* are accessing the network from an authenticated hospital device.
- **Anonymization and Masking:** When medical datasets are requested for research purposes, identity tools automatically mask patient identifying details, serving only clean, compliant, anonymous medical trends.

4. Completeness & Clarity

- **Conclusion:** Robust identity and access governance ensures that data accessibility never compromises a patient's fundamental right to medical record privacy.

Question 5: Virtualization & Hypervisors Supporting Scalability

1. Relevance to Application

- **Application Scenario:** Modern cloud SaaS or e-commerce platforms that experience dynamic resource requirements fluctuating daily, requiring optimal underlying hardware usage.
- **Relevance:** The hypervisor acts as the fundamental translation layer that allows multiple independent operating systems to seamlessly split a single set of physical resources.

2. Use of Cloud Concepts

- **Resource Overcommit & Consolidation:** Maximizing the efficiency of underlying CPU, memory, and storage allocation by assigning dormant capacity to active workloads.
- **Elastic Horizontal Scalability:** Spinning up or tearing down identical VM instances dynamically based on immediate processing loads.
- **Hardware Abstraction:** Decoupling software dependencies from specific physical hardware models, standardizing core deployments.

3. Practical Insight

- **Achieving Resource Efficiency:** Without hypervisors, a dedicated bare-metal server often runs at a mere 10%–15% CPU utilization capacity. Hypervisors consolidate dozens of disparate VMs onto that single server, pushing resource utilization to 80%+ safely.
- **Rapid Scale-Out Mechanisms:** During heavy data processing jobs, the hypervisor clones active production instances instantly from pre-configured master templates, utilizing shared storage repositories to accommodate sudden data analysis rushes without purchasing additional hardware.

4. Completeness & Clarity

- **Conclusion:** Hypervisors serve as the structural backbone of cloud computing, transforming static hardware limitations into elastic, cost-efficient, and highly scalable resources.

